



In [9]: `!pip install gensim`

```
Requirement already satisfied: gensim in /usr/local/lib/python3.12/dist-packages (4.4.0)
Requirement already satisfied: numpy>=1.18.5 in /usr/local/lib/python3.12/dist-packages (from gensim) (2.0.2)
Requirement already satisfied: scipy>=1.7.0 in /usr/local/lib/python3.12/dist-packages (from gensim) (1.16.3)
Requirement already satisfied: smart_open>=1.8.1 in /usr/local/lib/python3.12/dist-packages (from gensim) (7.6.0)
Requirement already satisfied: wrapt in /usr/local/lib/python3.12/dist-packages (from smart_open>=1.8.1->gensim) (2.1.2)
```

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```
In [8]: from gensim.models import Word2Vec
sentences = [
    ["king", "queen", "man", "woman"],
    ["paris", "france", "rome", "italy"],
    ["apple", "orange", "banana", "fruit"],
    ["car", "bus", "train", "vehicle"],
    ["man", "king", "boy"],
    ["woman", "girl"]
]

model = Word2Vec(sentences, vector_size=50, window=3, min_count=1, workers=2)
similarity = model.wv.similarity("king", "queen")
print("Similarity between king and queen:", similarity)
# Find most similar words
similar_words = model.wv.most_similar("king", topn=3)
print("Words similar to 'king':")
for word, score in similar_words:
    print(word, score)
# Word analogy example
result = model.wv.most_similar(positive=["king", "woman"], negative=["man"],
print("king- man + woman", result)
```

Similarity between king and queen: -0.057522435

Words similar to 'king':

boy 0.16554395854473114

orange 0.15518240630626678

france 0.14386031031608582

king- man + woman [('orange', 0.24333752691745758)]